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SEAT No. :

PD-451

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**B.E. (Honours in A.I. & M.L.)(Insem.)**  
**SOFT COMPUTING & DEEP LEARNING**  
**(2019 Pattern) (Semester - VIII) (410303)**

**Time : 1Hour]**

**[Max. Marks : 30**

**Instructions to the candidates:**

- 1) **Answer Q.1 or Q.2 and Q.3 or Q.4**
- 2) **Neat diagrams must be drawn wherever necessary**
- 3) **Figures to the right indicate full marks.**
- 4) **Make suitable assumptions wherever necessary.**

- Q1) a)** How do Hard Computing and Soft Computing differ in terms of characteristics, problem-solving approach and real-world application?[7]
- b)** Why is Soft Computing essential in modern problem-solving and how does it outperform traditional computing methods? [8]

OR

- Q2) a)** Discuss the major areas of Soft Computing and their real-world applications. [7]
- b)** Describe the evolution of computing and how it led to the development of Soft Computing. [8]

OR

- Q3) a)** Explain the Fuzzy Set Theory and how it differs from a Crisp Set. [7]
- b)** Describe Fuzzification and Defuzzification techniques with real-world examples. [8]
- Q4) a)** Explain Fuzzy Rule-Based Systems and their applications in decision-making. [7]
- b)** How are Fuzzy Classification techniques used in machine learning? Provide a case study or real-life example. [8]

